

Class 12 Timer Interrupt Task Program

```
/*  
 * File:  main.c  
 * Author: Sagar More  
 *  
 * Created on 19 June, 2025, 9:01 PM  
 */  
  
#include <xc.h>  
  
#include <stdint.h>  
  
  
#define _XTAL_FREQ 4000000  
  
// -- [[ Configuration Bits ]] --  
  
#pragma config FOSC = HS    // High-speed external oscillator (4-20 MHz crystal)  
#pragma config WDTE = OFF   // Watchdog Timer Enable (WDT disabled)  
#pragma config PWRTE = ON   // Power-up Timer Enable (enabled)  
#pragma config BOREN = ON   // Brown-out Reset Enable (enabled)  
#pragma config LVP = OFF    // Low-Voltage Programming (disabled)  
#pragma config CPD = OFF    // Data EEPROM Memory Code Protection (disabled)  
#pragma config WRT = OFF    // Flash Program Memory Write Enable (off)  
#pragma config CP = OFF     // Flash Program Memory Code Protection (disabled)  
  
uint8_t C = 0; // Global Counter Variable  
  
void timer1Init(void);
```

```
void main()
{
    // -- [[ IO Configurations ]] --
    TRISB = 0x00; // PORTB as output
    TRISD = 0x00; // PORTD as output (used in ISR)

    PORTB = 0x00;
    PORTD = 0x00;

    timer1Init();

    while(1)
    {
        PORTB = 0x00;
        __delay_ms(500);
        PORTB = 0xFF;
        __delay_ms(500);
    }
}
```

```

void timer1Init(void)
{
    // Choose the local clock source (timer mode) => TMR1CS = 0
    T1CON &= ~(0x01 << 1);

    // Choose the desired prescaler ratio (1:1) => T1CKPS1:T1CKPS0 = 00
    T1CON &= ~(0x01 << 4);
    T1CON &= ~(0x01 << 5);

    // Preload the value to TMR1 register
    TMR1 = 15535;

    // Clear the Timer1 overflow flag
    PIR1 &= ~(0x01 << 0);

    // Enable Timer1 overflow interrupt
    PIE1 |= (0x01 << 0);

    // Enable Peripheral and Global interrupts
    INTCON |= (0x01 << 6); // PEIE
    INTCON |= (0x01 << 7); // GIE

    // Turn ON Timer1 => TMR1ON = 1
    T1CON |= (0x01 << 0);
}

```

```

// Interrupt Service Routine - ISR
void __interrupt() ISR(void)
{
    // Check Timer1 overflow interrupt flag
    if (PIR1 & (0x01 << 0))
    {
        C++;
        if(C == 20)
        {
            PORTD = ~PORTD; // Toggle PORTD
            C = 0;
        }

        // Preload the value again
        TMR1 = 15535;

        // Clear the Timer1 interrupt flag
        PIR1 &= ~(0x01 << 0);
    }
}

```